5.1: Sequences

Definition.
A **sequence** is a function whose domain is either all the integers between two given integers or all the integers greater than or equal to a given integer.

Example. Given

$$a_k = \frac{k}{k+1},\ k \in \mathbb{Z}^{\geq 1} \qquad\qquad b_i = \frac{i-1}{i},\ i \in \mathbb{Z}^{\geq 2}$$

Compute the first five terms of both sequences

$a_1 = \frac{1}{1+1} = \frac{1}{2}$ $\qquad$ $b_2 = \frac{2-1}{2} = \frac{1}{2}$

$a_2 = \frac{2}{2+1} = \frac{2}{3}$ $\qquad$ $b_3 = \frac{3-1}{3} = \frac{2}{3}$

$a_3 = \frac{3}{3+1} = \frac{3}{4}$ $\qquad$ $b_4 = \frac{4-1}{4} = \frac{3}{4}$

$a_4 = \frac{4}{4+1} = \frac{4}{5}$ $\qquad$ $b_5 = \frac{5-1}{5} = \frac{4}{5}$

$a_5 = \frac{5}{5+1} = \frac{5}{6}$ $\qquad$ $b_6 = \frac{6-1}{6} = \frac{5}{6}$

Example. Find an explicit formula for a sequence with the following initial terms:

$$1,\ -\frac{1}{4},\ \frac{1}{9},\ -\frac{1}{16},\ \frac{1}{25},\ -\frac{1}{36},\ \ldots$$

$$\frac{(-1)^0}{1^2},\ \frac{(-1)^1}{2^2},\ \frac{(-1)^2}{3^2},\ \frac{(-1)^3}{4^2},\ \frac{(-1)^4}{5^2},\ \frac{(-1)^5}{6^2}$$

$$\Rightarrow a_k = \frac{(-1)^k}{(k+1)^2},\quad k \geq 0$$

Definition.
Let $m, n \in \mathbb{Z}$ with $m \leq n$. The **summation** of a_k from m to n is given by:

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_{n-1} + a_n$$

where k is the **index**, m is the **lower limit**, and n is the **upper limit**.

Example. Rewrite the following using summation notation:

$$1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \frac{1}{25} - \frac{1}{36} = \sum_{k=0}^{5} \frac{(-1)^k}{(k+1)^2}$$

Example. Compute $\sum_{k=1}^{5} k^2$ $= 1^2 + 2^2 + 3^2 + 4^2 + 5^2$

$= 1 + 4 + 9 + 16 + 25$

$= 55$

Example. Write $\sum_{k=1}^{n} 2^k + 2^{n+1}$ as a single summation

$$\sum_{k=1}^{n} 2^k + 2^{n+1} = \underbrace{2^1 + 2^2 + \dots + 2^{n-1} + 2^n}_{} + 2^{n+1} = \sum_{k=1}^{n+1} 2^k$$

Example. Rewrite $\sum_{i=1}^{n+1} \frac{1}{i^2}$ by separating off the final term

$$\sum_{i=1}^{n+1} \frac{1}{i^2} = \left(\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2}\right) + \frac{1}{(n+1)^2}$$

$$= \sum_{i=1}^{n} \frac{1}{i^2} + \frac{1}{(n+1)^2}$$

Example. Evaluate the following telescoping sum:

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1}\right)$$

$$\sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1}\right) = \left(\frac{1}{2} - \cancel{\frac{1}{3}}\right) + \left(\cancel{\frac{1}{3}} - \cancel{\frac{1}{4}}\right) + \cancel{\dots} + \left(\cancel{\frac{1}{n}} - \frac{1}{n+1}\right) = \boxed{\frac{1}{2} - \frac{1}{n+1}}$$

Definition.
Let $m, n \in \mathbb{Z}$ with $m \leq n$. The **product** of a_k from m to n is given by:

$$\prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdot \cdots \cdot a_{n-1} \cdot a_n$$

where k is the **index**, m is the **lower limit**, and n is the **upper limit**.

Example. Compute the following:

$$\prod_{k=1}^{5} k = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5$$

$$= 5!$$

$$= \boxed{120}$$

$$\prod_{k=1}^{10} \frac{k}{k+1} = \frac{1}{\cancel{2}} \cdot \frac{\cancel{2}}{\cancel{3}} \cdot \frac{\cancel{3}}{\cancel{4}} \cdot \cdots \frac{\cancel{10}}{11} = \boxed{\frac{1}{11}}$$

Example. Rewrite the following summation so that the index begins at $k = 1$:

$$\sum_{k=0}^{6} \frac{1}{k+1}$$

$$\sum_{k=0}^{6} \frac{1}{k+1} = \frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{7} = \sum_{j=1}^{7} \frac{1}{j}$$

$j = k+1$

$k=0 \Rightarrow j=1$

Definition.
For each positive integer n, the quantity n **factorial**, denoted $n!$, is defined to be the product of all of the integers from 1 to n:

$$n! = n \cdot (n-1) \ldots 3 \cdot 2 \cdot 1 = \prod_{k=1}^{n} k$$

Zero factorial, denoted $0!$, is defined to be 1:

$$0! = 1$$

Example. Simplify the following expressions:

$$\frac{8!}{7!} = \frac{8 \cdot \cancel{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}}{\cancel{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}} = \boxed{8}$$

$$\frac{5!}{2! \cdot 3!} = \frac{5 \cdot 4 \cdot \cancel{3!}}{2! \cdot \cancel{3!}} = \frac{5 \cdot 4}{2} = 5 \cdot 2 = \boxed{10}$$

$$\frac{(n+1)!}{n!} = \frac{(n+1) \cdot n!}{n!} = \boxed{n+1}$$

$$\frac{n!}{(n-3)!} = \frac{n \cdot (n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}} = \boxed{n \cdot (n-1)(n-2)}$$

$$\left(\frac{3}{3}\right)\frac{1}{2! \cdot \underbrace{4!}_{4 \cdot 3!}} + \frac{1}{\underbrace{3!}_{3 \cdot 2!} \cdot 3!}\left(\frac{4}{4}\right) = \frac{3}{3!\,4!} + \frac{4}{3!\,4!} = \boxed{\frac{7}{3!\,4!}}$$

Definition.
Let $n, r \in \mathbb{Z}$ with $0 \leq r \leq n$. Then

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

is n **choose** r, which represents the number of subsets of size r that can be chosen from a set with n elements.